Edward Ajayi

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Education

Carnegie Mellon University
Master of Science in Engineering Artificial Intelligence — *Aug. 2024 – Present*
- **Relevant Courses:** Natural Language Processing, Deep Learning, Question Answering with LLMs
- **Specializations:** Data Science, Machine Learning, NLP
- **GPA:** 3.77/4.0

Ekiti State University
Bachelor of Science in Computer Science — *Oct. 2014 – Nov. 2019*
- **Specializations:** Computer Science
- **GPA:** 4.17/5.0

Research Interests

Language Modeling, Mechanistic Interpretability, Multimodal Reasoning, Human-Centered AI

Selected Publications

1. **E. Ajayi**, M. Kachweka, M. Deku, and E. Aiken. *A Machine Learning Approach for Detection of Mental Health Conditions and Cyberbullying from Social Media. Proceedings of Machine Learning Research (PMLR), AAAI-26 AI in Medicine and Healthcare Bridge*. **Best Paper Award**. 2026. [PMLR]
2. **E. Ajayi** and P. Mitra. *HumorGen: Cognitive Synergy for LLM Humor Generation via Persona-Based Distillation*. Submitted to *EMNLP*. 2026. [arXiv Preprint]
3. **E. Ajayi** and P. Mitra. *HumorRank: An Automated Leaderboard for Evaluating Humor Generation in Large Language Models*. Submitted to *COLM*. 2026. [Manuscript]
4. **E. Ajayi** and P. Mitra. *Automatic Humor Detection: From Theoretical Foundations to Large Language Models*. Under Review. 2025. [Manuscript]
5. **E. Ajayi**, B. Tadele, E. Umwari, M. Deku, P. Singadi, C. Edeh, and J. Udahemuka. *Language Diversity: Evaluating AI Performance on African Languages*. 2025. [arXiv Preprint]
6. **E. Ajayi**. *AfriEconQA: A Benchmark Dataset for African Economic Analysis Based on World Bank Reports*. Under Review. 2026. [arXiv Preprint]

Experience

Predoctoral Research Fellow
Human-Algorithm Interaction Lab, Oxford Saïd Business School, University of Oxford, UK — *Feb. 2026 – Present*
- Conduct research on Human-AI collaboration, decision-making, and AI adoption in service systems at the Human-Algorithm Interaction Lab.

Research Assistant
Spatial and Language Technologies Lab, CMU — *Aug. 2025 – Present*
- Conducting research on NLP for automatic humour detection and generation, dataset preparation, model development, and evaluation.
- Contributing to projects on low-resource language technologies and advancing cultural understanding in large language models.

Graduate Researcher
Carnegie Mellon University — *Jan. 2025 – Present*
- Researched machine learning methods for detecting mental health conditions and cyberbullying from social media, resulting in a paper accepted at AAAI-26.
- Conducting two ongoing research projects in Question Answering with Large Language Models, focusing on reasoning, evaluation, and robustness.

Research Intern - Business Automation
Carnegie Mellon University — *Jun. 2025 – Aug. 2025*
- Developing and prototyping an agentic AI framework to automate complex business workflows.
- Researching and implementing AI safety protocols and robust LLM reasoning for ethical alignment and deployment.
- Analyzing agent performance to establish benchmarks for human-in-the-loop oversight and develop explainability metrics to support transparent corporate accountability.

Client Service Officer
Stanbic IBTC Bank — *Dec. 2021 – Jul. 2024*
- Leveraged data-driven insights to cross-sell over N400M in assets, enhancing client trust and digital adoption by 25%.
- Managed client relationships, identifying service gaps to improve operational efficiency.

Technical Skills

Languages: Python, SQL, Bash
Generative AI & LLMs: LangGraph, OpenAI Agents SDK, RAG, LLM Fine-Tuning, Prompt Engineering, DPO
Deep Learning & ML: PyTorch, Hugging Face Transformers, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn
Natural Language Processing: spaCy, NLTK, Sentiment Analysis, Text Processing
Data Engineering & Ops: Docker, AWS, FastAPI, Neo4j, BeautifulSoup, Jupyter

Selected Projects

HumorRank — LLM Evaluation & Ranking Framework | *Python, Evaluation* Leaderboard [Paper]
- Built an evaluation pipeline for ranking outputs from multiple LLMs on subjective generation tasks.
- Implemented pairwise comparison with probabilistic ranking using the Bradley-Terry model.
- Designed a candidate generation → evaluation → leaderboard workflow for scalable benchmarking.

HumorGen-7B — LLM-based Humor Generation Model | *PyTorch, Transformers* HuggingFace
- Developed HumorGen-7B, a fine-tuned 7B language model for controlled humor generation.
- Implemented end-to-end alignment pipeline using supervised fine-tuning and Direct Preference Optimization (DPO).
- Designed cognitive persona prompting framework to induce diverse reasoning paths for humor generation.

MyTorch — Minimal Deep Learning Framework Built from Scratch | *Python*
- Implemented core tensor operations, automatic differentiation, activation layers, and SGD without deep learning libraries.
- Designed a PyTorch-style API enabling end-to-end model training while illustrating neural network internals.
- Built modular components for educational experimentation with deep learning systems.

Mental Health & Cyberbullying Detection System (AAAI Oral) | *Transformers* GitHub [arXiv]
- Built an end-to-end multi-label classification system for detecting mental health signals and cyberbullying from social media data.
- Combined classical ML methods with transformer embeddings for robust prediction on noisy datasets.
- Produced results leading to an Oral paper acceptance at AAAI-26 AIMedHealth.

AI Workflow Automation Agent | *Python, LangChain, FastAPI*
- Built an autonomous agent to process submissions, route tasks, and trigger approval workflows.
- Integrated API orchestration, notifications, and real-time validation pipelines.

Training & Conferences

Oral & Poster Presentation, AAAI Conference on Artificial Intelligence — Singapore
AAAI — *Jan. 2026*
- Presented my paper on ML for mental health and cyberbullying detection from social media, and engaged with academics and researchers at the conference.

Participant, Deep Learning Indaba 2025 — Rwanda
Deep Learning Indaba — *Aug. 2025*
- Attended workshops on deep learning, AI for social good, and research collaboration across African universities.

Participant, Digital Humanities Summer School — Rwanda
DIGIHUM/CMU-Africa — *Jul. 2025*
- Engaged in computational methods for humanities research, including NLP applications.

Volunteering/Leadership

President, CMU-Africa Research Club
Carnegie Mellon University — *Jan. 2025 – Dec. 2025*
- Led 12 major events in 2025 supporting student research.
- Launched website with research guides, conference updates, and collaboration hub.
- Organized first Student Research Showcase with 20+ posters and workshops.
- Hosted three-part PhD Application Series, faculty sessions, and alumni panels supporting 40+ students.
- Collaborated with ACM SIGCOMM and Upanzi Network for remote conference participation.
- Built an active research ecosystem that provided critical peer support, contributing to over 20 student acceptances to top-tier venues including NeurIPS and AAAI in 2025.

Reviewer, AAAI AIMedHealth Bridge 2026
AI for Medicine and Healthcare — *Nov. 2025 – Nov 2025*
- Reviewed submissions in applied machine learning and AI for healthcare under the AAAI AIMedHealth Bridge 2026 program.

IEEE Student Member, CMU-Africa
Institute of Electrical and Electronics Engineers — *Feb. 2025 – Present*
- Collaborating with IEEE members to advance research initiatives among club members.

Certifications

- Microsoft Certified Azure Data Scientist Associate - View
- DataCamp Python Data Associate - View

Honors and Awards

- Mastercard Foundation Scholarship, Carnegie Mellon University Africa
- Federal Scholarship Board Award for Undergraduates, Ekiti State University